Image Alignment : Part 3

→ Goals ① How to establish constraints for a calibrated camera used to capture Images for the purpose of creating a panorama of a given scene.

② How to create a 360 panoramic view

Knowing how to perform Change of coordinates: Cartesian & spherical ones.

First Goal: Assume we have two cameras P_1 & P_2

$$P_1 = K_1 R_1 [I | -\tilde{C}_1] = K_1 [R_1 | \vec{t}_1]$$

$$P_2 = K_2 R_2 [I | -\tilde{C}_2] = K_2 [R_2 | \vec{t}_2]$$

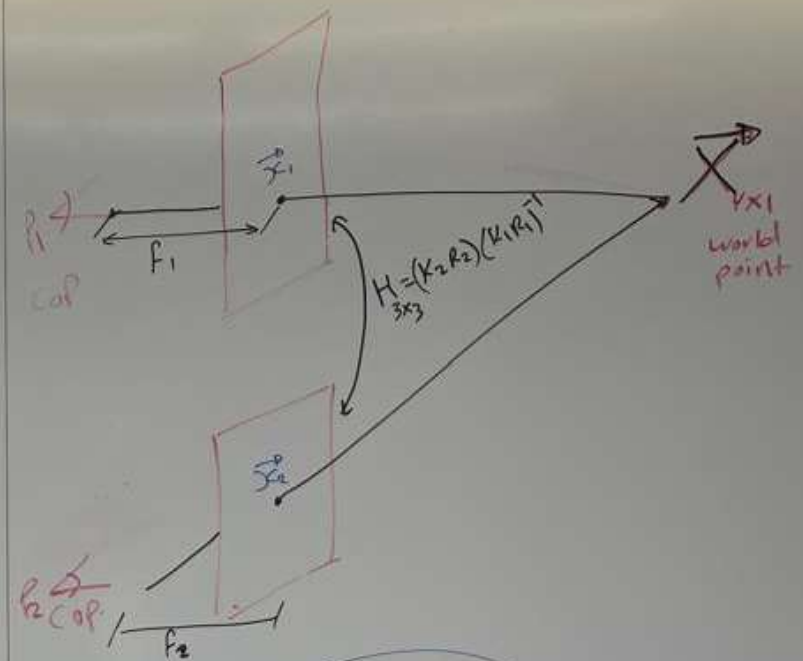
Now, if we impose that P_1 & P_2 CoPs are the same, when K_s' & R_s' are allowed to change, we can have the following constraint between P_1 & P_2.

$$P_2 = (K_2 R_2)(K_1 R_1)^{-1} P_1$$ **fact**

↳ only change we will study

Case ① Changing only Rotations while fixing K_s'.

Case ② Changing only focal Length while fixing R_s'.



$$\underset{3\times1}{\vec{x}_1} = P_1 \underset{4\times1}{\vec{X}}$$

$$\underset{3\times1}{\vec{x}_2} = P_2 \underset{4\times1}{\vec{X}} = (K_2 R_2)(K_1 R_1)^{-1} P_1 \underset{4\times1}{\vec{X}} = \underset{3\times3}{(K_2 R_2)(K_1 R_1)^{-1}} \underset{3\times1}{\vec{x}_1}$$

Substitute

$$\underset{3\times1}{\vec{x}_2} = \underbrace{(K_2 R_2)(K_1 R_1)^{-1}}_{3\times3 \text{ non-singular}} \underset{3\times1}{\vec{x}_1}$$

$$\downarrow$$

$$H_{3\times3}$$

identity.

Case 1 Let $R_1 = I$, $R_2 = R$ & $K_1 = K_2 = K$

then $H_{3\times3} = (KR)(KI)^{-1} = KRK^{-1}$

Case 2 Let $R_1 = R_2 = \text{Identity} = I$

then $H = (K_2 I)(K_1 I)^{-1} = K_2 K_1^{-1}$

$\text{Eigenvalues}(R) \equiv \text{eigenvalues of } (H)$

Conjugate Rotation matrix

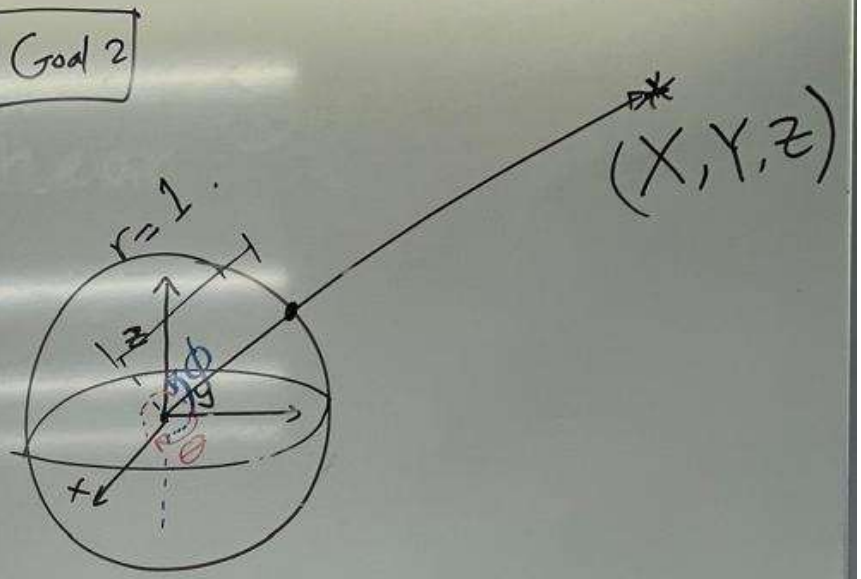
↓ perform eigen decomposition

$\theta = \cos^{-1}\left(\frac{\lambda_2 + \lambda_3}{2}\right)$

3 eigenvalues

$\lambda_1 = 1,\ \lambda_2 = e^{i\theta},\ \lambda_3 = e^{-i\theta},\ i = \sqrt{-1}$

Goal 2



sphere.

(r, θ, ϕ)

θ: $(0 - 2\pi)$

ϕ: $(0 - \pi)$

1

unit sphere.

$(\hat{x}, \hat{y}, \hat{z}) =$